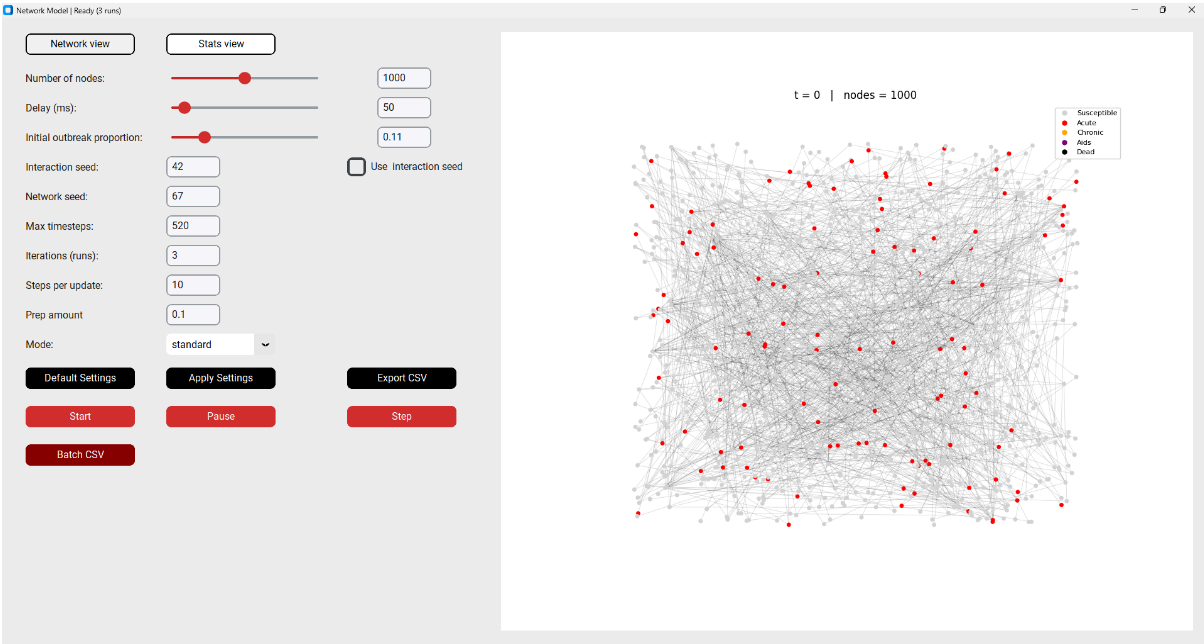
Strategizing HIV Prevention

A Stochastic Model of HIV Transmission Dynamics in Heterogeneous Sexual Networks for optimized PrEP Distribution.

Team members: Chaja Bakker, Anne Beks, Jelmer van Dam, Glenn de Graaff
Team number and team name (optional): Team 5

URL of github repository:
https://github.com/SystemExit/Computational-Science-Project



Brief summary of the work:

The HIV epidemic remains a global health crisis with universal Pre-Exposure Prophylaxis (PrEP) treatment being economically and practically unfeasible. This project identifies which subpopulation should be targeted to minimize HIV incidence. We hypothesized that targeting bisexual bridges would yield the most reduction in total infections. To test this, we built a stochastic network model based on a sexual contact network derived from the Egodyads dataset. We performed simulations for six socio-sexual classes, accounting for biological multipliers and social PrEP-intake dynamics. The results showed that targeting the male homosexual subgroup outperformed the random distribution at identical coverage levels. These findings provide a framework for the optimisation of HIV interventions under budget constraints.

Details of the work and the results:

Model design

Data from HIV transmission network studies that reported the gender, HIV status and which nodes (persons) had sex with whom, was used to create a synthetic sexual network of a 1000 nodes with a similar degree distribution [1]. The network was initialised by attributing to each node the properties gender and sexuality, representative of global prevalence and infecting the same proportion of nodes in the synthetic network as in the original HIV transmission network [2]. Hereby the nodes could be divided into six subgroups corresponding to their gender and sexuality, such that all nodes in the same subgroups were assigned the same probability of having sex per week.

After the initialisation, sexual transmission of HIV was simulated by passing 520 weeks (10 years) time, and letting the nodes have sex with their neighbours probabilistically based on the probability of having sex given their sexual orientation and gender, and normalised by dividing by the node's degree. The sexual contact may or may not conclude with the transmission of HIV, based on if the two nodes have HIV, whether they used condoms and if the condoms worked, and whether or not the nodes used antiretroviral therapy (ART) or PrEP. When a node is newly infected, the node is in the acute phase with high viral load and therefore a higher probability of transmitting HIV than in the chronic phase that follows it, which is characterised by low viral load and few symptoms. From the start of infection it takes 12 weeks to go into the chronic phase and 442 weeks (8.5 years) to develop AIDS if the HIV infection is left untreated. However, an infected individual node most likely has symptoms of an HIV infection that may or may not warrant them to be screened for HIV. If they do get screened, the HIV infection is assumed to be diagnosed, and the person will take ART, which permits them to have a normal life expectancy and avoid developing AIDS. If they do not get screened the person will develop AIDS and have a higher probability of transmitting HIV. Nodes with AIDS are assumed to immediately take ART to increase their life expectancy, which still decays roughly exponentially with time.

Results

Targeted PrEP provision to male homosexuals resulted in the highest number of susceptible individuals after 10 years. Providing PrEP to 30 male homosexuals yielded 856 susceptible individuals, whereas more than 200 individuals required PrEP when distributed at random or only to heterosexuals to achieve comparable results (Appendix A.2). Sensitivity analysis (Appendix A.1) showed that outcomes were most influenced by transmission probabilities, while testing and condom use had moderate effects.

Conclusion

Targeting PrEP to male homosexuals is the most effective and cost-efficient strategy to reduce HIV transmission. Future research should incorporate dynamic network structures with forming and dissolving connections to better reflect sexual behaviour and improve policy relevance.

Team member contribution:

Anne Beks

- Peer review 1
- Created network model code
- Plotting code adjustments
- Final code coherence check
- Final report

Chaja Bakker

- Plotting code
- Literature research (parameters)
- Poster creation
- Developed and contributed the largest part to the project idea.
- Final report

Glenn de Graaff

- Peer review 1
- Created GUI code
- Plotting code adjustments
- Data generation code
- Final report

Jelmer van Dam

- Peer review 2
- Literature research (network creation)
- R code for calculating the life-expectancy of AIDS patients that do not use ART.
- Network creation code
- Sensitivity analysis code
- Final report

Code contributions (instead of git-fame):

Author	File	Commits
Glenn de Graaff	visuals.py	Made visuals.py (660 loc)
Anne Beks	network_model.py	Made network_model.py (307 loc)
Jelmer van Dam	create_sexual_network.py	Made create_sexual_network function (262 loc)
Anne Beks	create_sexual_network.py	Fixed logical errors, added unit tests and added the sexual_frequency function (153 loc)

Jelmer van Dam	sensitivity_analysis.py	Created sensitivity analysis function and plots (226 loc)
Glenn de Graaff and Anne Beks	sensitivity_analysis.py	Adjusted plots for visual appeal and correct labeling
Chaja Bakker	plots.py	Created plots.py (83 loc)
Chaja Bakker	best_prep_plots.py	Created best_prep_plots.py (17 loc)
Chaja Bakker	plots_people.py	Created plots_people.py (93 loc)
Jelmer van Dam	Survival_rate_with_AIDS_after_t_years.R	Created fit of survival rate with AIDS after t years (23 loc)

References

1. Morris, M., & Rothenberg, R. (2011). HIV transmission network metastudy project: An archive of data from eight network studies, # 1988--2001.
2. Rahman, Q., Xu, Y., Lippa, R. A., & Vasey, P. L. (2020). Prevalence of Sexual Orientation Across 28 Nations and Its Association with Gender Equality, Economic Development, and Individualism. Archives of sexual behavior, 49(2), 595–606.

References not directly cited in this report, but used in the creation of the model. Explanation is in the ReadMe:

Boily, M. C., Baggaley, R. F., Wang, L., Masse, B., White, R. G., Hayes, R. J., & Alary, M. (2009). Heterosexual risk of HIV-1 infection per sexual act: systematic review and meta-analysis of observational studies. *The Lancet. Infectious diseases*, *9*(2), 118–129. https://doi.org/10.1016/S1473-3099(09)70021-0

Cohen, M. S., Chen, Y. Q., McCauley, M., Gamble, T., Hosseinipour, M. C., Kumarasamy, N., Hakim, J. G., Kumwenda, J., Grinsztejn, B., Pilotto, J. H. S., Godbole, S. V., Mehendale, S., Chariyalertsak, S., Santos, B. R., Mayer, K. H., Hoffman, I. F., Eshleman, S. H., Piwowar-Manning, E., Wang, L., Makhema, J., … Fleming, T. R. (2011). Prevention of HIV-1 infection with early antiretroviral therapy. *The New England Journal of Medicine, 365*(6), 493–505. https://doi.org/10.1056/NEJMoa1105243

Hollingsworth, T. D., Anderson, R. M., & Fraser, C. (2008). HIV-1 transmission, by stage of infection. *The Journal of infectious diseases*, *198*(5), 687–693. https://doi.org/10.1086/590501

McCormack, S., Dunn, D. T., Desai, M., Dolling, D. I., Gafos, M., Gilson, R., Sullivan, A. K., Clarke, A., Reeves, I., Schembri, G., Mackie, N., Bowman, C., Lacey, C. J., Apea, V., Brady, M., Fox, J., Taylor, S., Antonucci, S., Khoo, S. H., Rooney, J., … Gill, O. N. (2016). Pre-exposure prophylaxis to prevent the acquisition of HIV-1 infection (PROUD): effectiveness results from the pilot phase of a pragmatic open-label randomised trial. *Lancet (London, England)*, *387*(10013), 53–60. https://doi.org/10.1016/S0140-6736(15)00056-2

Mercer, C. H., Tanton, C., Prah, P., Erens, B., Sonnenberg, P., Clifton, S., Field, N., Macdowall, W., Lewis, R., Datta, J., Copas, A. J., Phelps, A., Wellings, K., & Johnson, A. M. (2013). Changes in sexual attitudes and lifestyles in Britain through the life course and over time: Findings from the National Surveys of Sexual Attitudes and Lifestyles (Natsal-3). *The Lancet, 382*(9907), 1781–1794. https://doi.org/10.1016/S0140-6736(13)62035-8

Morgan, D., Mahe, C., Mayanja, B., Okongo, J. M., Lubega, R., & Whitworth, J. A. (2002). HIV-1 infection in rural Africa: is there a difference in median time to AIDS and survival compared with that in industrialized countries?. *AIDS (London, England)*, *16*(4), 597–603. https://doi.org/10.1097/00002030-200203080-00011

Patel, P., Borkowf, C. B., Brooks, J. T., Lasry, A., Lansky, A., & Mermin, J. (2014). Estimating per-act HIV transmission risk: a systematic review. *AIDS (London, England)*, *28*(10), 1509–1519. https://doi.org/10.1097/QAD.0000000000000298

Raiteri, R., Fora, R., & Sinicco, A. (1994). No HIV-1 transmission through lesbian sex. *Lancet (London, England)*, *344*(8917), 270. https://doi.org/10.1016/s0140-6736(94)93037-6

Time from HIV-1 seroconversion to AIDS and death before widespread use of highly-active antiretroviral therapy: a collaborative re-analysis. Collaborative Group on AIDS Incubation and HIV Survival including the CASCADE EU Concerted Action. Concerted Action on SeroConversion to AIDS and Death in Europe. (2000). *Lancet (London, England)*, *355*(9210), 1131–1137.

Appendix

A. Figures

A.1: Sensitivity analysis

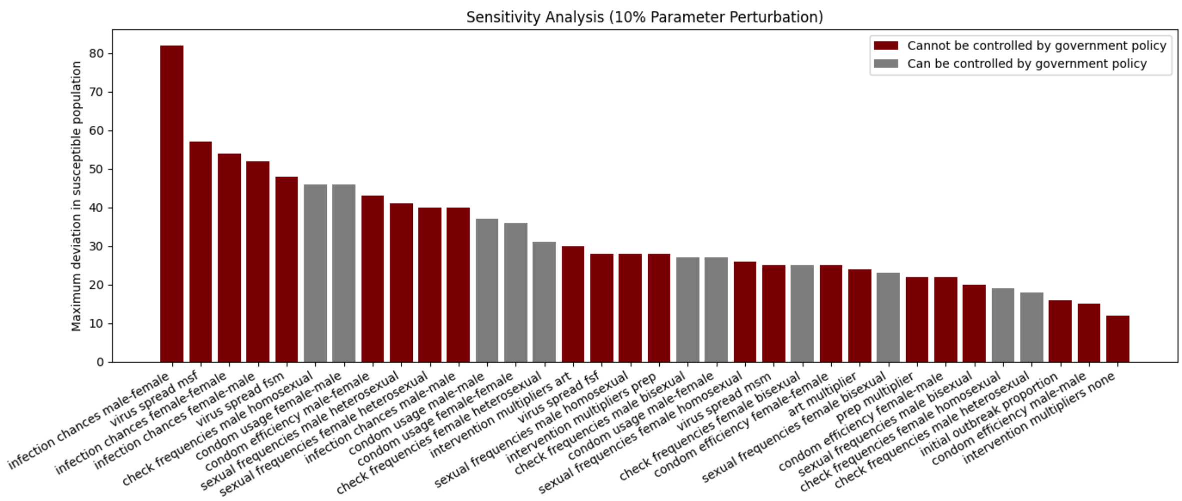


Figure 1. Deviation in susceptible population based on 10% parameter perturbation sorted from highest influence to lowest.

A.2: Results PrEP effectiveness

Divided into three plots based on the x-axis scale

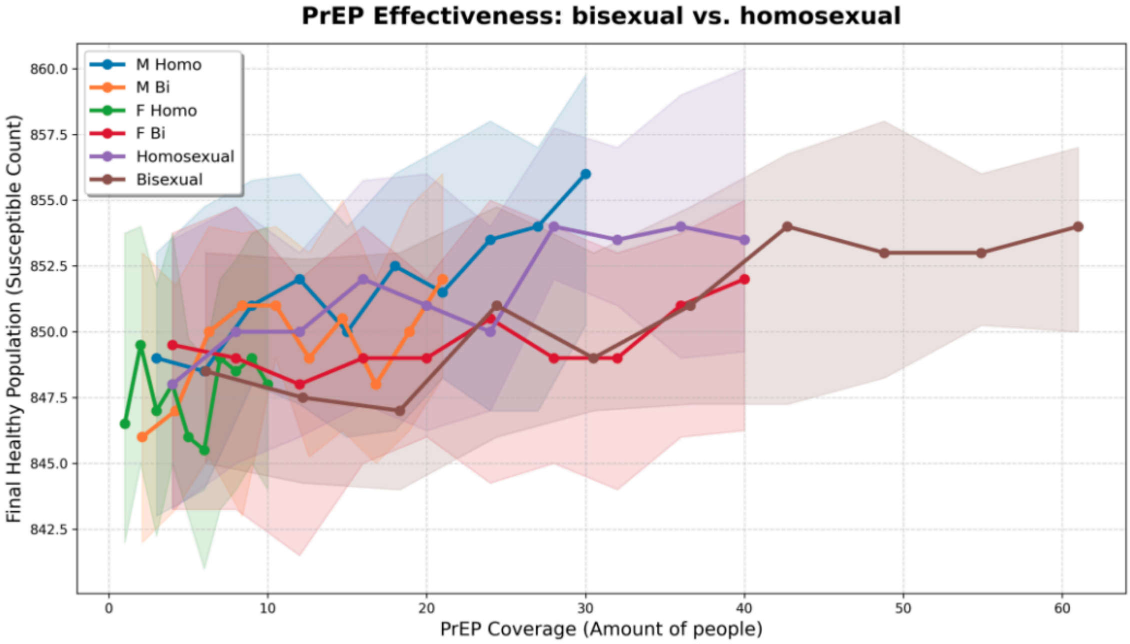


Figure 2. Amount of susceptible people as a function of the proportion of the individuals of the group that were randomly assigned to take PrEP over 10 years.

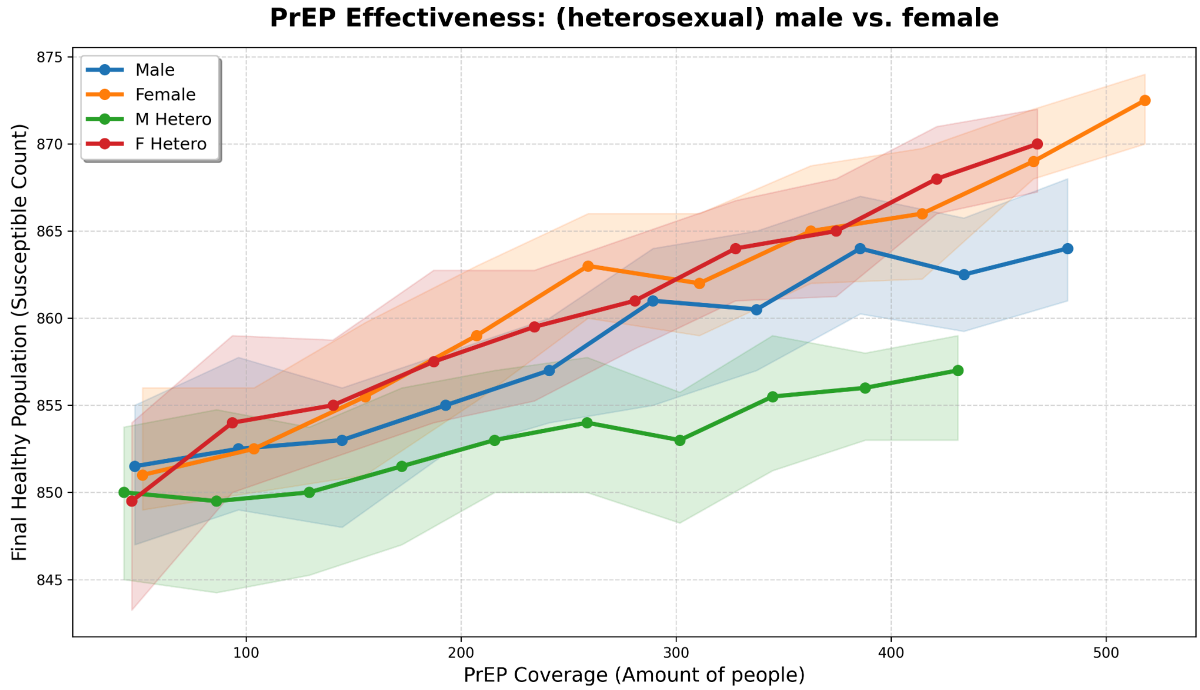


Figure 3. Amount of susceptible people as a function of the proportion of the individuals of the group that were randomly assigned to take PrEP over 10 years.

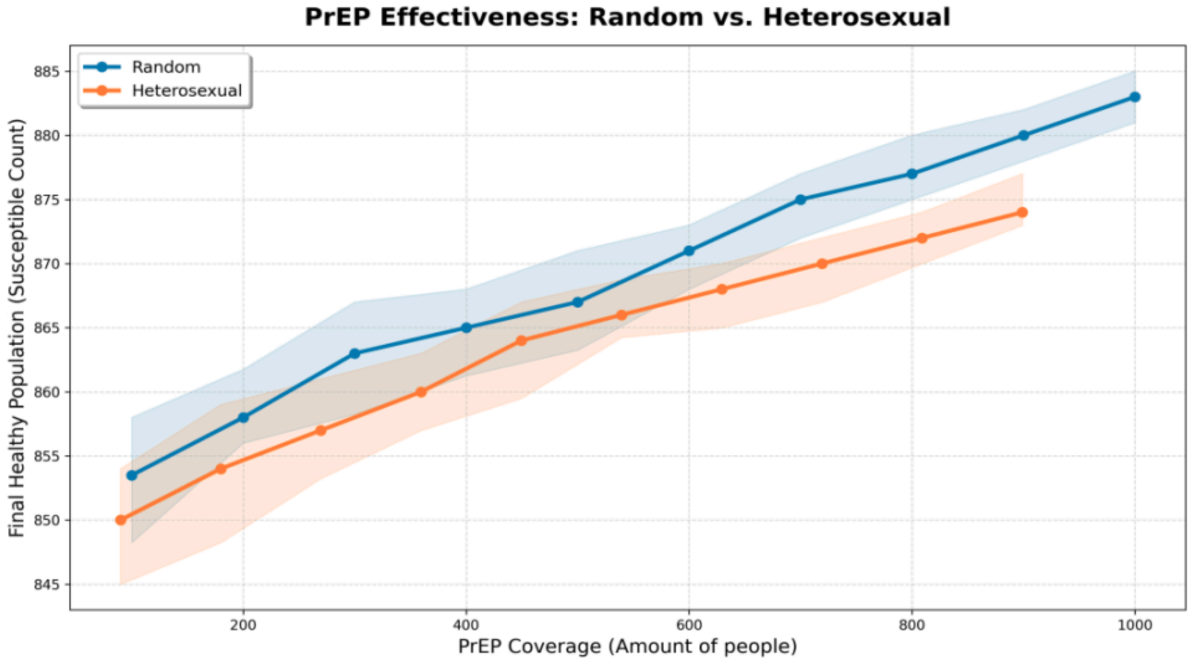


Figure 4. Amount of susceptible people as a function of the proportion of the individuals of the group that were randomly assigned to take PrEP over 10 years.

B: Peer review

Project proposal peer review 1

Submitted by team: Chaja Bakker, Anne Beks, Jelmer van Dam, Glenn de Graaff
To review the project titled: Modeling the Spatial Spread of Neuronal Degeneration in Parkinson's Disease Using Cellular Automata
From team: Joris Berkenbosch, Louis Jansz, Lucas Nagtegaal and Wenyi Zhang

After reading the project proposal thoroughly, using the guidelines on what to review from the initial lecture slide titled "Peer review - Use the template from Canvas!", we make the following constructive remarks (min 50, max 200 words):

<u>Scientific question:</u>
Problem context: Use less jargon or better explain terms for a broader audience.
Research question: Research questions starting with 'To what extent…' require clear measurement guidelines to validate the model, such as defining metrics and thresholds for success or failure.

Hypothesis: The hypothesis doesn't seem to address the research question. It focuses on specific neuron behavior (likely in the CA model, but this is unclear) and its outcome won't directly conclude your research. It might be intended as a metric for model correctness, but this is not evident in the proposal.

<u>Numerical method:</u> You do not really describe the way that you intend to use the model. Comparing the model's outcome to existing data is a good validation method, but ensure metrics are used to quantify the prediction accuracy. Further, the model doesn't mention stochasticity. Disease spread is likely probabilistic, requiring the model to be stochastic. If so, run enough simulations to get a representative average result.

<u>Division of work:</u>The plan for division of work sounds nice in theory, with everyone contributing to the overall work and the responsibilities being distributed flexibly. However, in practice the absence of clear divisions of work made beforehand may lead to some people doing a lot of work while others lag behind. If this way of dividing tasks works for your team that's great, just be sure that it doesn't cause any problems.

<u>Timeline:</u> You are planning to read all literature and create the whole concept of the model in week 1, this doesn't seem feasible.

Project proposal peer review 2

Submitted by team: Chaja Bakker, Anne Beks, Jelmer van Dam, Glenn de Graaff
To review the project titled: Emergence of Structure in a Simplified Universe
From team: Toan Võ, Xintong Yao, Yael Ai

After reading the project proposal thoroughly, using the guidelines on what to review from the initial lecture slide titled *"Peer review - Use the template from Canvas!"*, we make the following constructive remarks (min 50, max 200 words):

The scientific question and objective of the project could be specified more specifically; for now it is not clear what particles the project aims to study and what is defined to be structure, clustering, and randomness. The research questions are not yet stated in such a fashion that they can be rejected empirically and the hypothesis is missing. We would suggest you restrict your project to a single particle system, e.g. crystallisation of certain atoms, for which empirical data is already available so your model can be validated. As a final remark, we suggest that the division of work is assigned to the team members explicitly, such that the division of workload is well-balanced, and members can be held accountable should they not perform their tasks.

Group 6:

- The research question is very optimistic. I'm unsure if you'll have enough time to handle all parts of the question, so you might need to revise it later. This can also be seen in the plan for division of labour; there are a lot of points to complete in the four weeks of the project. You could take another look at the timeline and division of work and discuss if it is a reasonable plan or if it’s too optimistic.

- The research question also seems a bit broad. You could try to change the “how” question into a more specific and easily answerable question. Also splitting it up into multiple sections could make it more comprehensible.

- The project proposal is missing a hypothesis, which would help with demonstrating that you have a reasonable understanding of how the network will behave.

- There are no datasets or papers linked in the proposal. It might be nice to have literature to fall back on and to support your project.

Group 7:
We believe the proposal is well-structured and clearly written. The research question is relevant and clearly formulated, and the choice of an agent-based model to study HIV transmission on heterogeneous sexual networks is well justified, especially given the limitations of homogeneous SIR-type models. The proposal demonstrates a solid understanding of both epidemiological theory and network science, and the planned comparison between heterogeneous and homogeneous models adds clear analytical value.

Another highlight of the proposal is the inclusion of targeted interventions, such as ART and PrEP, which makes the project highly policy-relevant. Moreover, the numerical method, tools, and validation strategy are described in sufficient detail, and the division of works is fairly clear and realistic.

We believe one possible improvement would be to specify which historical datasets will be used for validation and how goodness-of-fit will be assessed. Additionally, adding more critical thinking, such as briefly discussing potential limitations could further strengthen the proposal.

Overall, this is a strong and feasible project with clear scientific contribution.